

Q1 Find the minimum number in an array and its index.

arr = [⁰5, ¹7, ²2, ³1, ⁴12]

Ans. Algorithm

- i) let the 0-th element be the minimum.
- ii) Now, start a loop from 1 to the length of the array.
 - For each element compare if that element is lesser than your pre-existing minimum
 - If it is bigger than pre-existing minimum, then do nothing and continue.
 - If smaller, update your pre-existing minimum number with this.

Q1 Let's say, we are given a list of numbers and we want to arrange them in increasing order.

$num = [5, 7, 2, 1, 12]$

Ans $\rightarrow$ sorted-num = $[1, 2, 5, 7, 12]$

Way-1 (Storing in a new array)

i) Start a loop(j) starting from 0 to $\text{length}(num)$

- within each loop, find the minimum and its index.
* Exclude the indices present in already-sorted-ind

- Store the j-th minimum in the new array sorted-num.

$\text{sorted-num}[j] = \rightarrow$

- Store the index of the minimums already found in some other array called already-sorted-ind

Way-2 (Storing in the same array)

$\rightarrow$ Run a loop(i) on the index of the array from 0 to length of the array-1.

within that

$\rightarrow$ store the j-th element as the minimum.

$\rightarrow$ Try to find the ^{actual} minimum and its index by comparison (exactly same as previous,

$\rightarrow$ Exchange the starting minimum with new minimum.

[5, 7, 2, 1, 12]

j=0

$$\text{min_num} = \text{num}[j] \equiv \text{num}[0] = 5$$

$$\text{min_num_ind} = j \equiv 0$$

for i in range ():

⋮

we will find new min-num and new min-num-ind.

$$\text{min_num} = 1$$

$$\text{min_num_ind} = 3$$